

深度学习及应用



实践—卷积神经网络

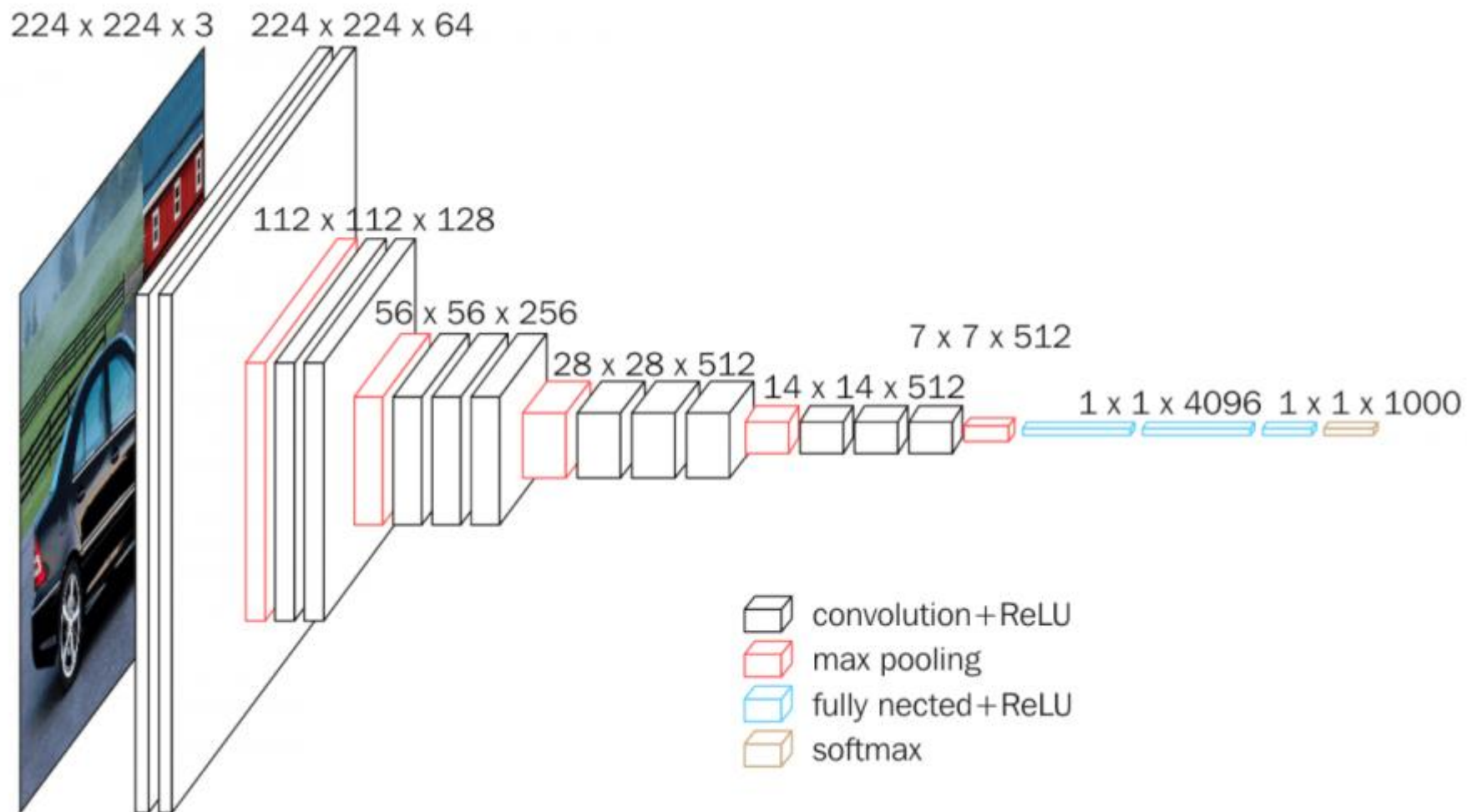
侯淇彬

南开大学计算机学院

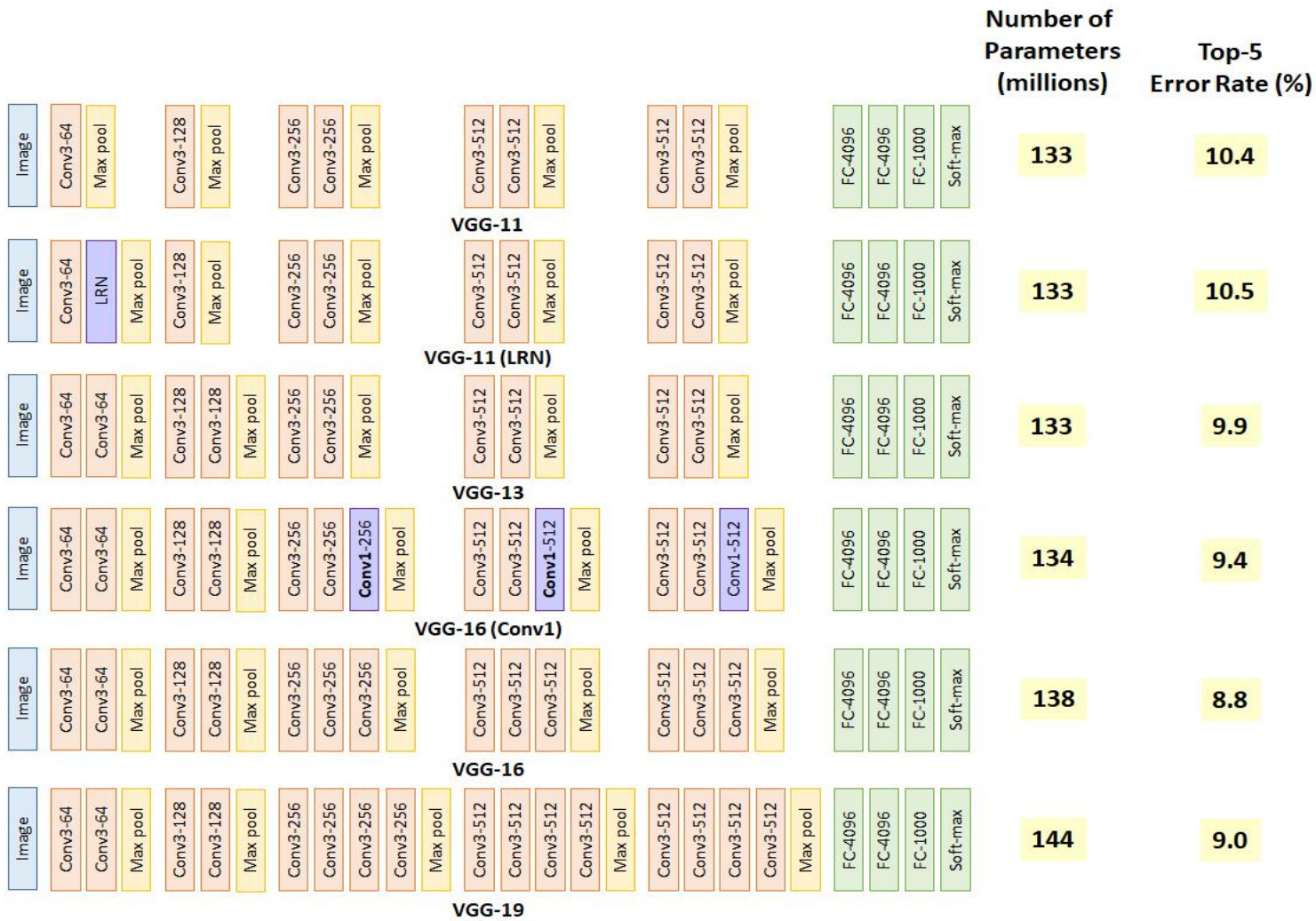
实验要求

- 掌握PyTorch框架基础算子
 - Convolution, Pooling, Activation layers
- 学会使用PyTorch搭建简单的卷积神经网络来训练Cifar10数据集
- 了解经典的卷积神经网络结构

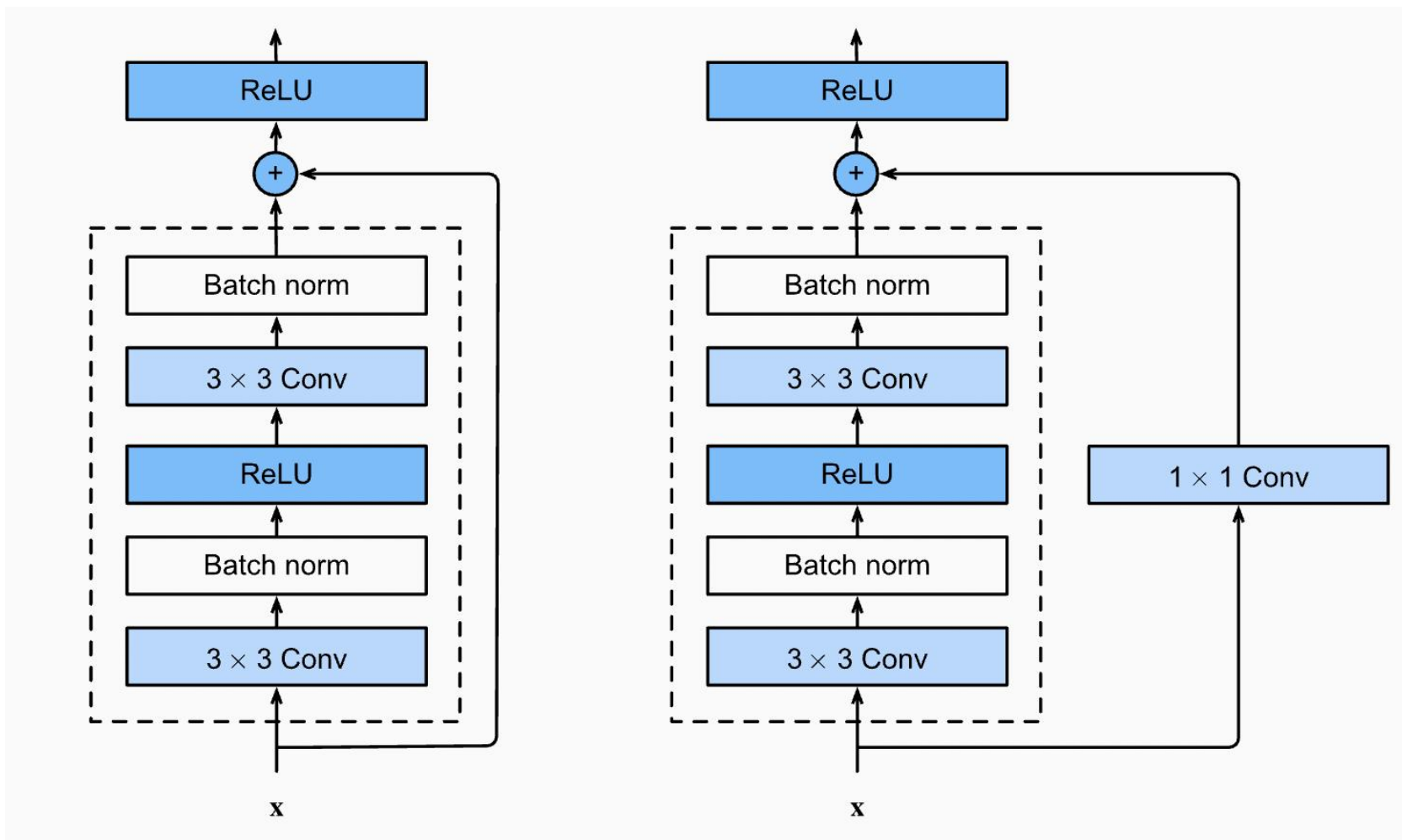
VGGNet



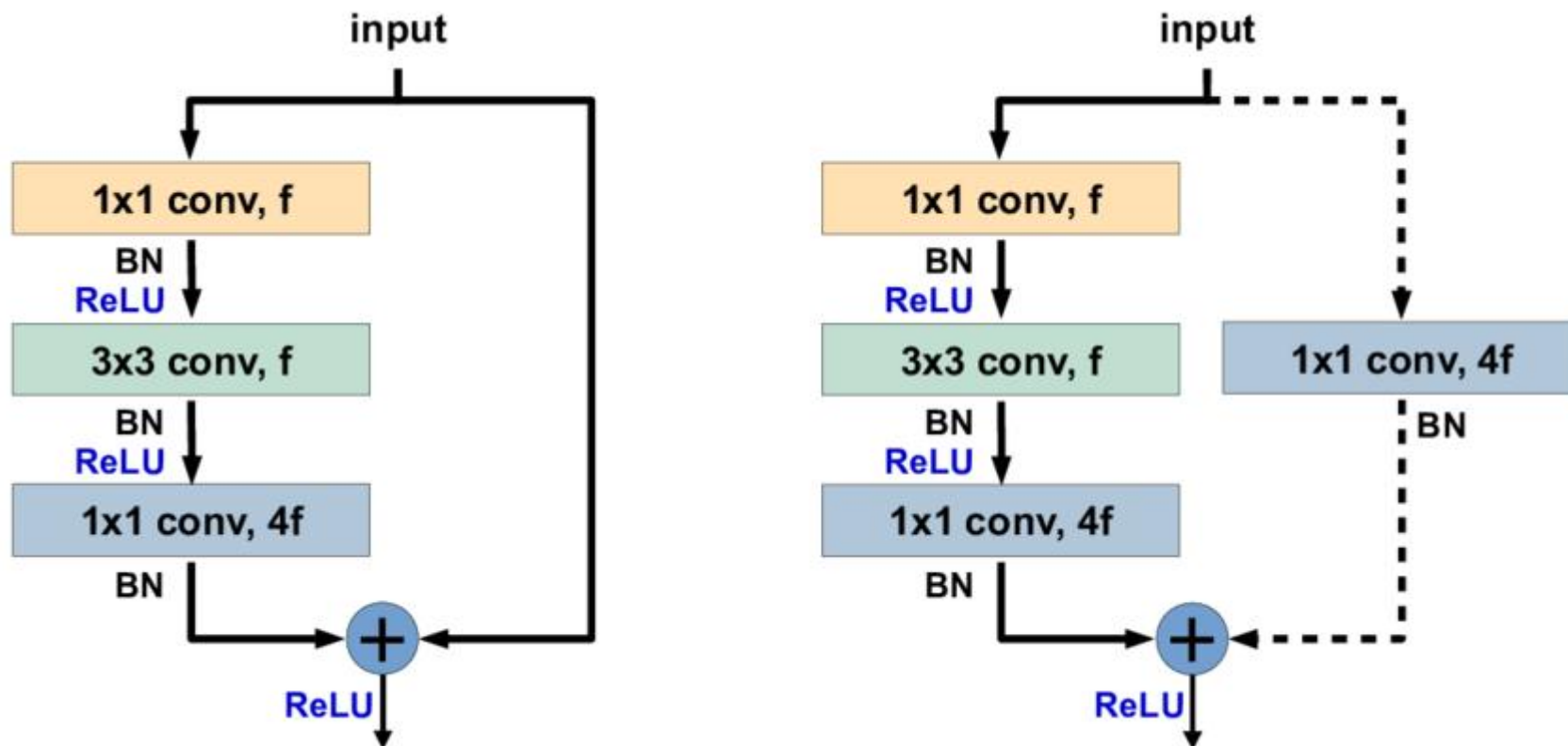
VGGNet



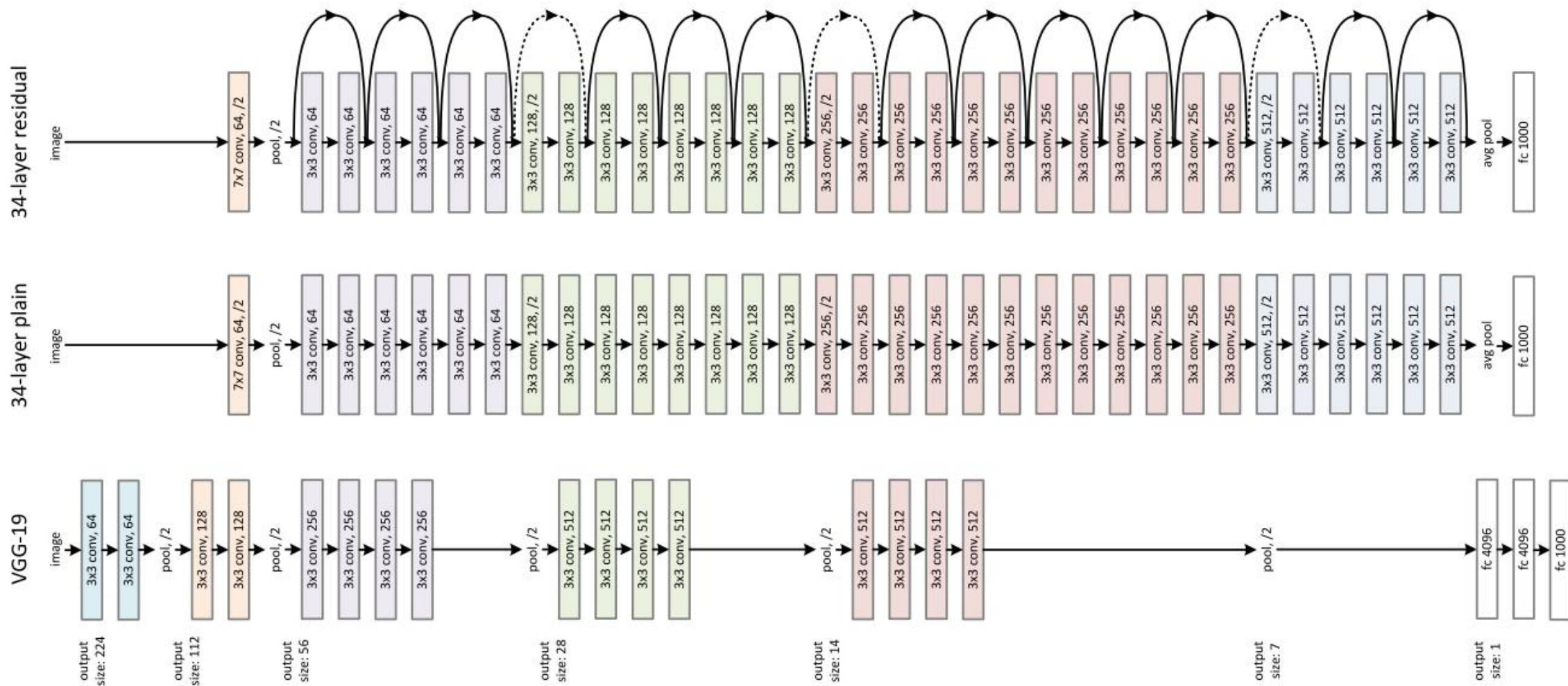
ResNet



ResNet



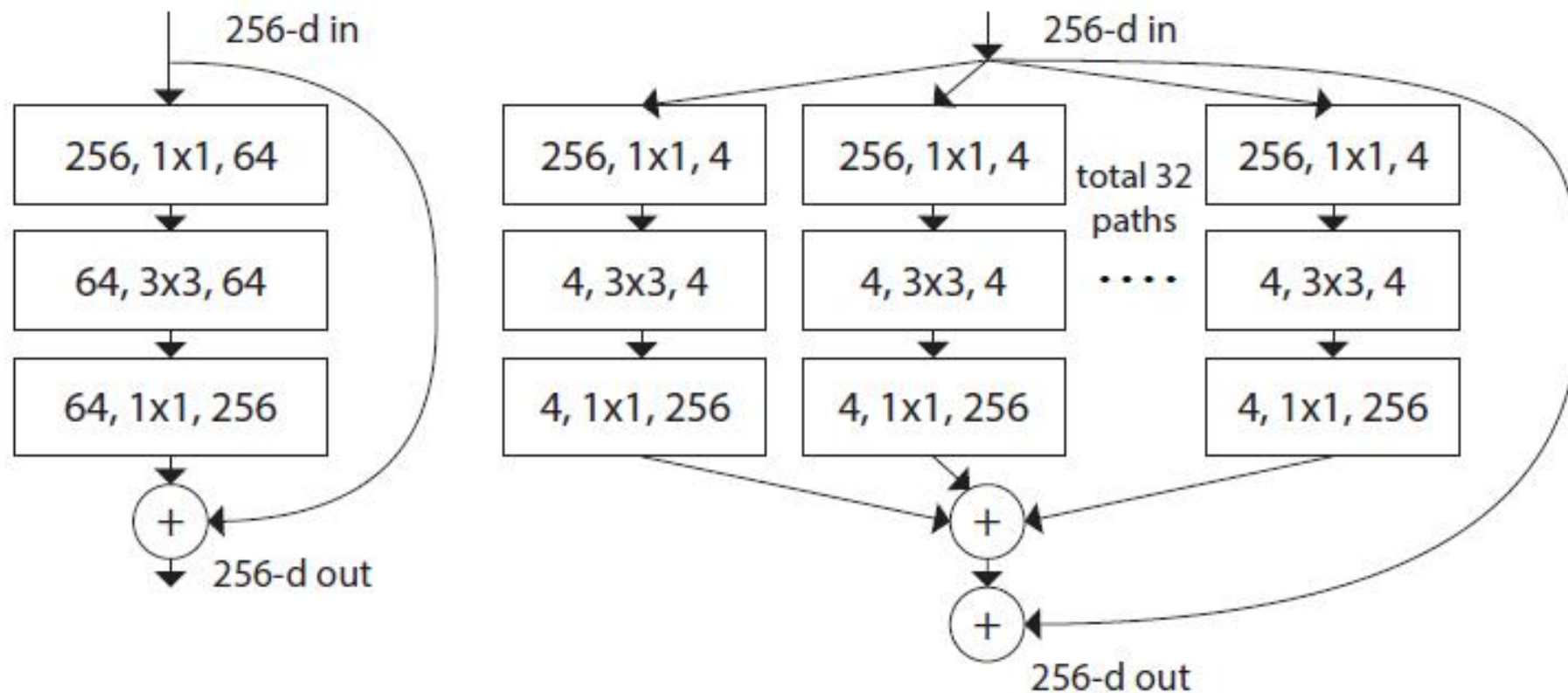
ResNet



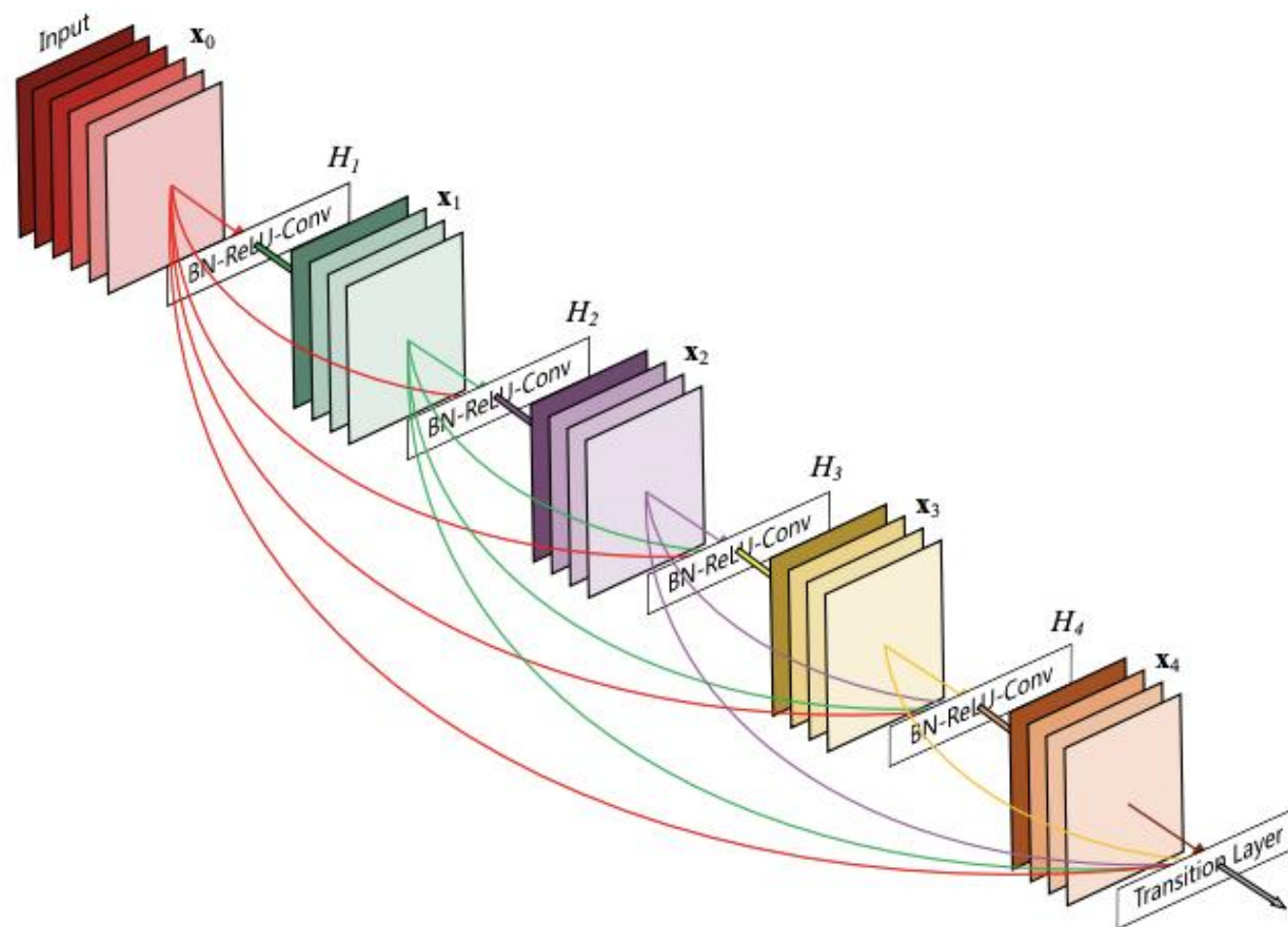
ResNet

layer name	output size	18-layer	34-layer	50-layer	101-layer	152-layer
conv1	112×112	$7 \times 7, 64, \text{stride } 2$				
conv2_x	56×56	$3 \times 3 \text{ max pool, stride } 2$				
		$\begin{bmatrix} 3 \times 3, 64 \\ 3 \times 3, 64 \end{bmatrix} \times 2$	$\begin{bmatrix} 3 \times 3, 64 \\ 3 \times 3, 64 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$
conv3_x	28×28	$\begin{bmatrix} 3 \times 3, 128 \\ 3 \times 3, 128 \end{bmatrix} \times 2$	$\begin{bmatrix} 3 \times 3, 128 \\ 3 \times 3, 128 \end{bmatrix} \times 4$	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 4$	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 4$	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 8$
conv4_x	14×14	$\begin{bmatrix} 3 \times 3, 256 \\ 3 \times 3, 256 \end{bmatrix} \times 2$	$\begin{bmatrix} 3 \times 3, 256 \\ 3 \times 3, 256 \end{bmatrix} \times 6$	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 6$	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 23$	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 36$
conv5_x	7×7	$\begin{bmatrix} 3 \times 3, 512 \\ 3 \times 3, 512 \end{bmatrix} \times 2$	$\begin{bmatrix} 3 \times 3, 512 \\ 3 \times 3, 512 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$
	1×1	average pool, 1000-d fc, softmax				
FLOPs		1.8×10^9	3.6×10^9	3.8×10^9	7.6×10^9	11.3×10^9

ResNeXt



DenseNet

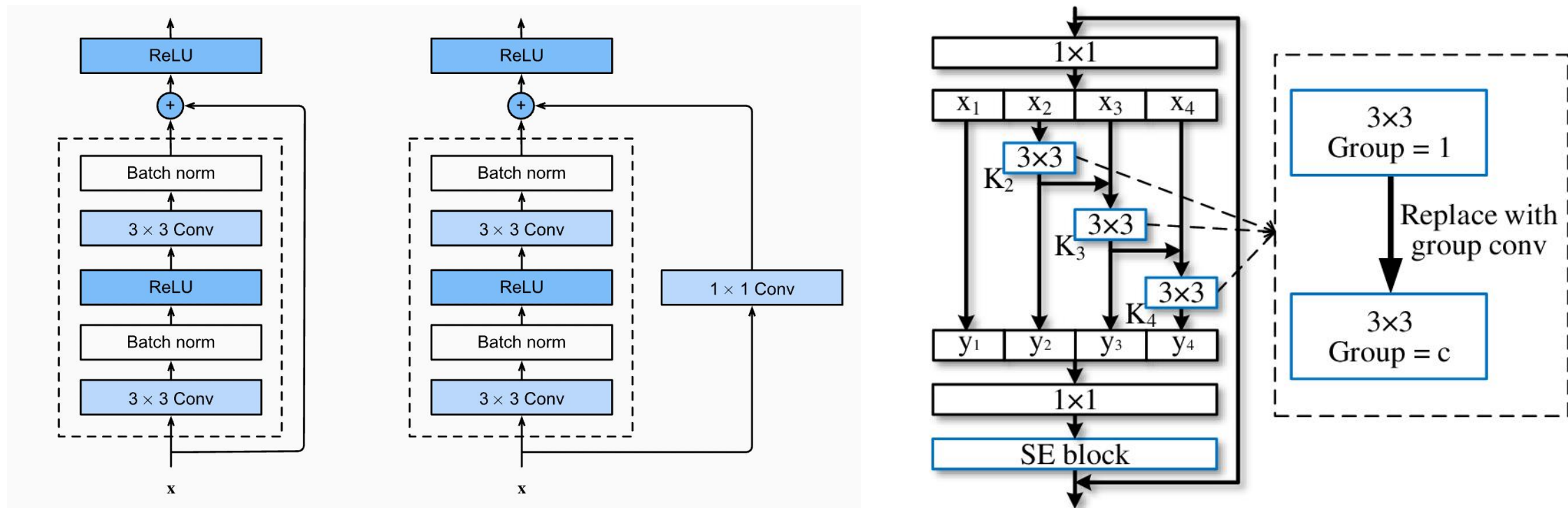


延伸阅读1

- **Res2Net** (Res2Net: A New Multi-scale Backbone Architecture)
- **VAN** (Visual Attention Networks)
- **MobileNets** (MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications)
- **MobileNetv2** (MobileNetV2: Inverted Residuals and Linear Bottlenecks)
- **MobileNeXt** (Rethinking Bottleneck Structure for Efficient Mobile Network Design)

扩展练习

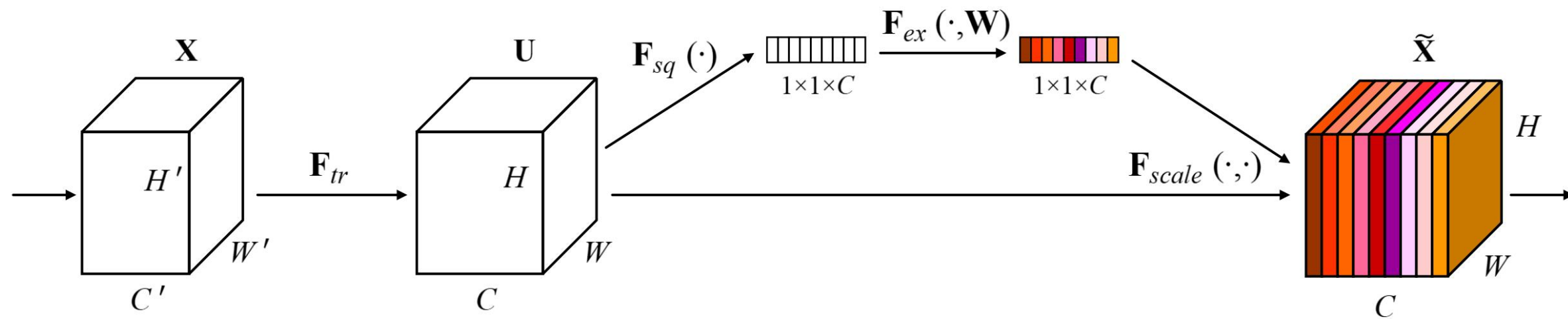
- **Res2Net** (Res2Net: A New Multi-scale Backbone Architecture)



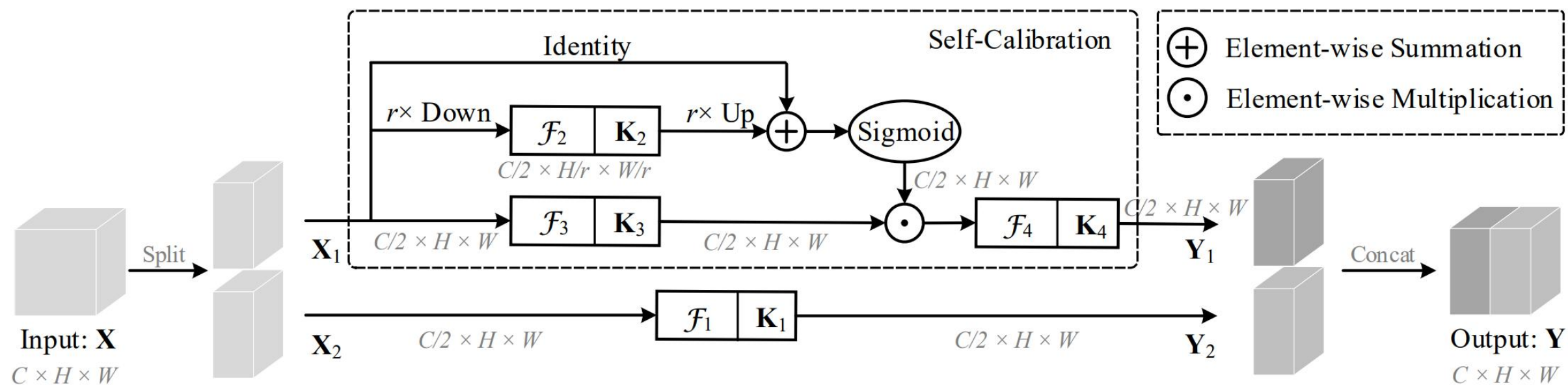
延伸阅读2

- **SENet** (Squeeze-and-excitation networks)
- **Self-Calibrated Convolution** (Improving Convolutional Networks with Self-Calibrated Convolutions)
- **SKNet** (Selective kernel networks)
- **Dynamic Convolution** (Dynamic Convolution: Attention over Convolution Kernels)

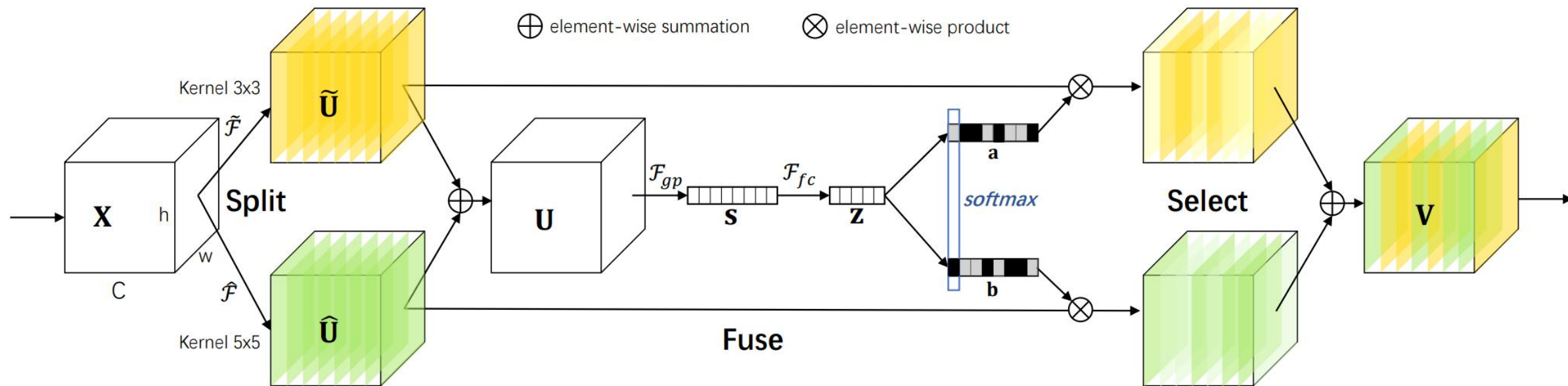
SENet



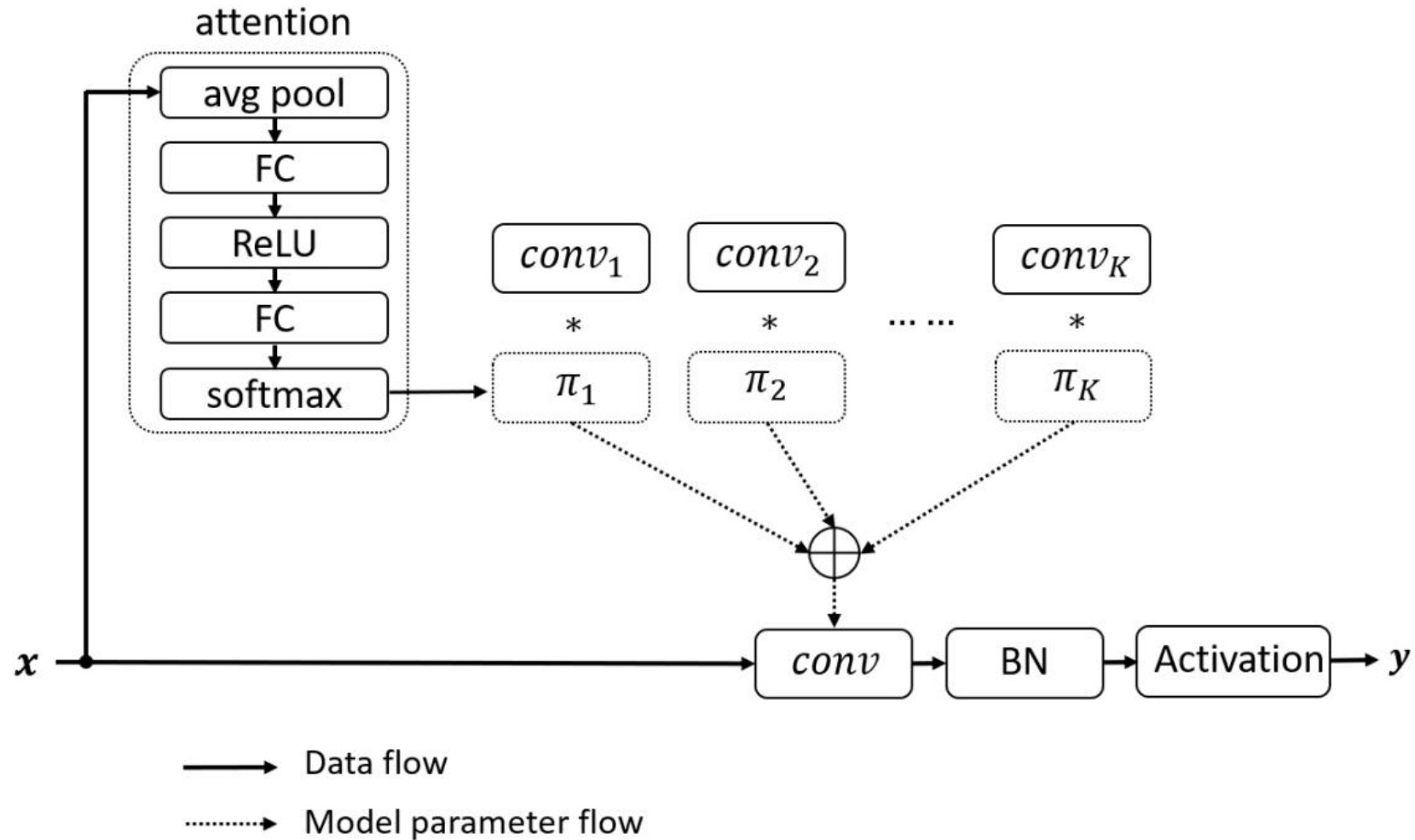
Self-Calibrated Convolution



SKNet



Dynamic Convolution



报告内容

- 老师提供的原始版本卷积网络结构（可用`print(net)`打印，复制文字或截图皆可）、在Cifar 10验证集上的训练loss曲线以及准确度曲线图
- 个人实现的微型ResNet网络结构、在Cifar 10验证集上的训练loss曲线以及准确度曲线图
- 个人实现的微型DenseNet网络结构、在Cifar 10验证集上的训练loss曲线以及准确度曲线图
- 个人实现的带有Depthwise conv的MobileNets网络结构、在Cifar 10验证集上的训练loss曲线以及准确度曲线图

作业提交

- 报告和代码（可将jupyter notebook里代码复制到一个xxx.py文件中）
打包（学号+姓名.zip），提交方式另行通知
- 实验报告内容应工整，**报告内容不超过8页**